

1 Calibration

confidence c_k ; context h_k
calibration label y_k
score $s_k = f(c_k, h_k)$

2 Risk gate (fixed $\mathcal{F} \times \mathcal{T}$)

$U_{CP} \leq \alpha$: admit K^+
otherwise: $K^+ = \emptyset$
fixed family; $N \geq n_{\min}$

y_k : calibration only; never deployed.

certified only

3a Data X

raw observations
 $X \in \mathbb{R}^{n \times p}$

3b Support $S(X)$

data-supported
pairs only
fixed before fusion

3c Permitted projection

data directions + certified K^+
orient $S(X)$; no new edges
exact DP: $p \leq 16$
relocation: $p > 16$

4 Audited DAG

support-preserving
and acyclic
not graph FDR / SHD

Invariant: $\text{supp}(A) \subseteq S(X)$ and A is acyclic